

PROJECT 12

Capstone Project: Cloud-Native AI Model Deployment Platform

Specialization: Cloud Native Software Engineering

Duration: 6 months (Capstone-1: 3 months | Capstone-2: 3 months)

Team Size: 4 Students

1. Title

Cloud-Native AI Model Deployment Platform

2. Aim

Create a cloud-native platform to deploy and manage machine learning models using containerized services.

3. Outcomes

Capstone-1 Outcomes (Months 1-3)

- Design cloud-native architecture and system components.
- Setup containerized development environment.
- Implement initial microservices and APIs.

Capstone-2 Outcomes (Months 4-6)

- Deploy services using container orchestration.
- Implement monitoring, logging, and scaling.
- Final system testing and documentation.

4. Hardware Components and Cost Breakdown

S.No	Component	Quantity	Cost (INR)
1	Raspberry Pi 4 (Test Node / Edge Client)	1	4000
2	MicroSD Card 64GB	1	600
3	Laptop / Development System	1	0 (Lab Resource)
4	External SSD	1	3500

5	WiFi Router / Network	1	1000
6	Cloud Platform (AWS/Azure/GCP Free Tier)	1	0
7	Cooling Fan	1	400
8	Cables & Accessories	1 set	800
TOTAL			9000 – 15000

4. Revised Hardware Components and Cost Breakdown

S.No	Component	Quantity	Cost (INR)
1	Raspberry Pi 4 (Test Node / Edge Client)	1	4000
2	MicroSD Card 64GB	1	600
3	Laptop / Development System	1	0 (Lab Resource)
4	External SSD	1	3500
5	WiFi Router / Network	1	1000
6	Cloud Platform (AWS/Azure/GCP Free Tier)	1	0
7	Cooling Fan	1	400
8	Cables & Accessories	1 set	800
TOTAL			9000 – 15000

Removed Item Details

Removed Item	Price (INR)	Reason
Dedicated Enterprise Server	₹50,000+	Cloud infrastructure used

		instead
Large Data Center Hardware	₹1,00,000+	Cloud provider resources utilized
High-End GPU Cluster	₹70,000+	Not required for cloud-native prototype

5. Team Composition and Role Distribution

Student 1: Cloud-Native Architect

Primary Responsibilities:

- Design microservices architecture and system topology.
- Define containerization strategy and service communication.

Skills Required: Microservices architecture, system design.

Student 2: DevOps and Containerization Engineer

Primary Responsibilities:

- Implement Docker containers and Kubernetes deployment.
- Configure CI/CD pipelines for automated deployment.

Skills Required: Docker, Kubernetes, DevOps tools.

Student 3: Backend Microservices Developer

Primary Responsibilities:

- Develop backend APIs and microservices.
- Implement service communication and data storage.

Skills Required: Backend development, REST APIs, databases.

Student 4: System Integration and Project Lead

Primary Responsibilities:

- Integrate microservices, infrastructure, and deployment pipelines.
- Perform testing, documentation, and final demonstration.

Skills Required: System integration, testing, documentation.